The Celestial Sphere: A Map for the Sky

What you will be able to do

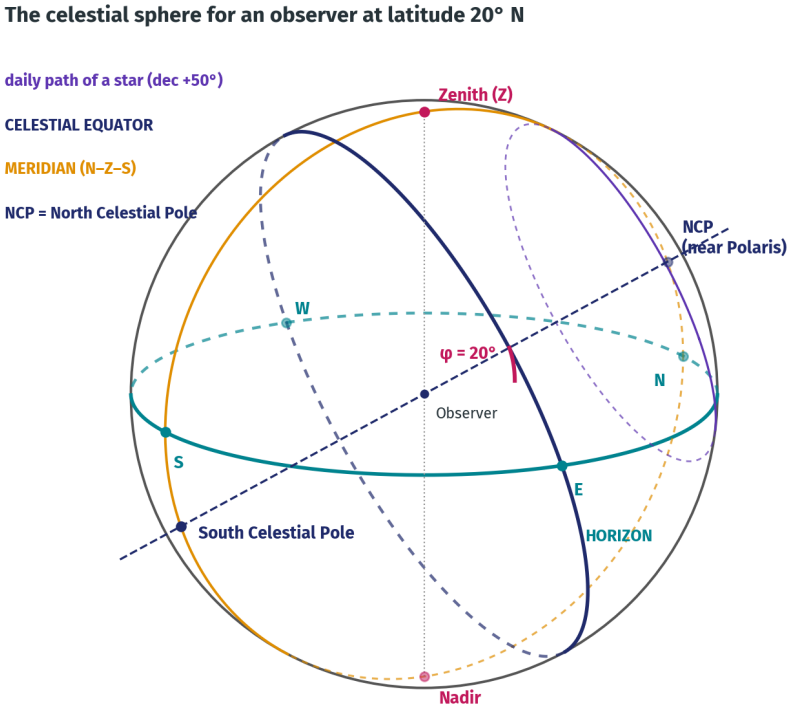
1. Describe the **celestial sphere** model and why it is useful even though it is not "real".
2. Define and point to: **zenith, nadir, horizon, cardinal points, meridian, prime vertical, celestial poles, celestial equator, ecliptic, equinox and solstice points**.
3. Use the rule **"altitude of the celestial pole = latitude"** to decide which stars never set and which never rise.
4. Measure angles in the sky with your hand and with the small-angle formula.

1. The model: stars painted on a giant ball

Stars lie at very different distances — Sirius is 8.6 light-years away, Deneb about 2,600 — but all are so far that our eyes cannot judge their distances. So it is very useful to **imagine every object painted on the inside of an enormous sphere**, with Earth at its centre: the **celestial sphere**.

- It seems to **turn once a day** (really, Earth turns — Lecture 1.1).
- At any moment you see only **half** of it; the other half is below your horizon.
- On this sphere we care only about **directions and angles**, not distances.

Astronomers in Greece, India and China all used this model — and modern astronomers still do: telescope mounts, star atlases and apps such as Stellarium are built on it.



The celestial sphere for an observer at latitude 20° N (e.g. Mumbai or Nagpur). Dashed lines lie on the far side.

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P.T.O.

2. Your personal reference points (they move with you)

Term	Meaning	Memory hook
Zenith	the point directly **overhead**	straight up along a plumb line
Nadir	the point directly **below your feet**	the zenith's "twin"
Horizon	the great circle 90° from the zenith, dividing the visible half of the sky from the hidden half	the rim of an upturned bowl
Cardinal points	N, E, S, W on the horizon	*Never Eat Soggy Waffles* (clockwise on a map)
Meridian	great circle from the North point, through the zenith, to the South point	a star is highest when it crosses it (*transit*)
Prime vertical	great circle from East, through the zenith, to West	
Altitude	angle above the horizon (0°–90°)	fists from the horizon
Zenith distance	$z = 90° - \text{altitude}$	

These points belong to **you**: travel from Delhi to Chennai and your zenith moves to a different place among the stars.

3. The sky's own reference points (fixed among the stars)

Term	Meaning
North Celestial Pole (NCP)	the point above Earth's North Pole; the sky turns around it. **Polaris** is within 0.7° of it.
South Celestial Pole (SCP)	above Earth's South Pole; no bright star marks it; never visible from India.
Celestial equator	Earth's equator projected onto the sky, 90° from both poles. It always meets the horizon **exactly at E and W**.
Ecliptic	the Sun's yearly path, tilted 23.44° to the celestial equator (Lecture 1.2).
First Point of Aries	where the Sun crosses the celestial equator going north (∼20 March): the zero point of star coordinates.
Solstice points	the most northern (+23.44°) and southern (−23.44°) points of the ecliptic.
Diurnal circle	the circle a star traces in one day, parallel to the celestial equator.

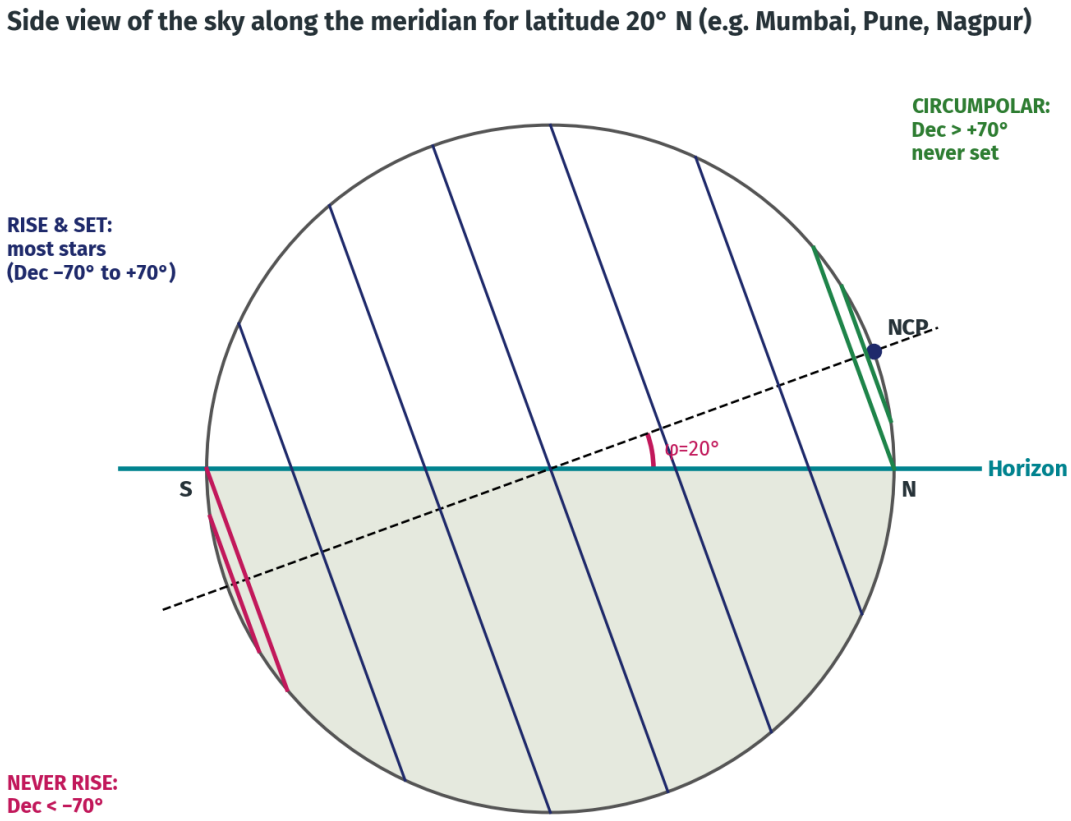
The most important rule in this unit

Altitude of the celestial pole = observer's latitude φ

- At the **North Pole** ($\varphi = 90°$) the pole is at the zenith; stars circle parallel to the horizon.
- At the **equator** ($\varphi = 0°$) both poles sit on the horizon; stars rise and set vertically.
- In **India** ($\varphi \approx 8°$–$35°$) the pole is low in the north and most stars rise at a steep slant.

It follows that the **celestial equator crosses the meridian at altitude** $90^\circ - \varphi$**, in the south** — 71° above the southern horizon in Mumbai.

4. Circumpolar stars: who never sets?



Each straight line is the edge-on daily circle of a star. Rule: circumpolar if Dec > 90° − φ; never seen if Dec < −(90° − φ). Delhi (28.6° N): circumpolar above +61.4°. At the Equator: none! At the North Pole: every visible star is circumpolar.

Side view along the meridian at latitude 20° N.

For a northern observer at latitude φ, a star of declination δ (its angle north or south of the celestial equator) is

circumpolar if $\delta > 90^\circ - \varphi$, never visible if $\delta < -(90^\circ - \varphi)$, otherwise it rises and sets.

Worked examples

Mumbai ($\varphi = 19^\circ$): $90^\circ - 19^\circ = 71^\circ$. Kochab ($\delta = +74^\circ$) is circumpolar; Dubhe ($+62^\circ$) is **not** — the Saptarishi sets in Mumbai! The Southern Cross (-57° to -63°) **can** be seen low in the south.

Delhi ($\varphi = 28.6^\circ$): $90^\circ - 28.6^\circ = 61.4^\circ$. Dubhe ($+61.8^\circ$) is just circumpolar, scraping the horizon; Acrux (-63.1°) is below -61.4°, so it never rises in Delhi.

India's latitude is a gift

From southern India you can see the Southern Cross, Alpha Centauri and Canopus — *and* Polaris. Few countries see so much of the sky.

5. Measuring angles in the sky

You met the **hand ruler** in Lecture 1.1 (little finger $\approx 1°$, three fingers $\approx 5°$, fist $\approx 10°$, spread hand $\approx 20°$–$25°$). Calibrate it on the sky:

- the three stars of **Orion's Belt** span about $3°$;
- the two **pointer stars** of the Saptarishi are $5.4°$ apart, and the whole Saptarishi is about $25°$ long;
- horizon to zenith is $90° = 9$ fists.

Units of angle

$$1 \text{ circle} = 360°, \qquad 1° = 60' \text{ (arcminutes)}, \qquad 1' = 60'' \text{ (arcseconds)}.$$

The Moon is about $30'$ wide; the sharpest human eye resolves about $1'$; Jupiter's disc is about $40''$–$50''$.

The small-angle formula

For small angles, an object of real size D at distance d has angular size

$$\theta \text{ (radians)} = \frac{D}{d} \qquad \text{or} \qquad \boxed{\theta\,(^\circ) = 57.3 \times \frac{D}{d}} \qquad (1 \text{ radian} = 57.3°).$$

Worked examples

The Moon: $\theta = 57.3 \times \dfrac{3474 \text{ km}}{384{,}400 \text{ km}} = 0.52°$ — about half a degree ✔

A 1-rupee coin ($D \approx 2.3\,\text{cm}$) just covers the Full Moon at a distance $d = 57.3 \times \dfrac{2.3 \text{ cm}}{0.52} \approx 250\,\text{cm} = 2.5\,\text{m}$. Try it!

Try tonight

1. Point to your **zenith**, then to the **N, S, E and W** points.
2. Trace the **meridian** with your arm: from the North point, over your head, down to the South point.
3. Find **Polaris** and measure its altitude — that is your latitude.
4. Trace the **celestial equator**: from the East point, up to $90° - \varphi$ in the south (≈ 7 fists in Mumbai) and down to the West point. The westernmost star of Orion's Belt (Mintaka) lies almost exactly on it.

Key points to remember

- Celestial sphere: an imaginary sphere on which we map the **directions** of objects.
- Zenith overhead, nadir below, horizon $90°$ from the zenith, meridian = N–zenith–S.
- **Altitude of NCP = latitude**; the celestial equator meets the horizon at E and W.
- Circumpolar if $\delta > 90° - \varphi$; never visible if $\delta < -(90° - \varphi)$.
- Small-angle formula: $\theta(^\circ) = 57.3\,D/d$.

Check yourself

1. At the North Pole, which celestial point coincides with your zenith?
2. Is Capella ($\delta = +46°$) circumpolar from Srinagar ($34°$ N)? From Bengaluru ($13°$ N)?
3. Can someone in Kanyakumari ($8°$ N) see Polaris? How high is it?
4. Two stars are 3 fists and 2 fingers apart. What is their approximate separation?

Answers: 1. The NCP. 2. Srinagar: $90 - 34 = 56$ and $46 < 56$, so no; Bengaluru: no. 3. Yes, but only $\sim 8°$ up — often lost in haze. 4. $\approx 32°$.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §2.1 "The Sky Above" — `https://openstax.org/books/astronomy-2e/pages/2-1-the-sky-above`
- **Stellarium**: switch on the equatorial grid, the meridian and the celestial equator (sky & viewing options → Markings) to see these circles on the real sky.